

RAY POMPONIO

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PROFESSIONAL SUMMARY

Biomedical Data Scientist with 7 years' experience specializing in statistical analysis and scientific research. Proven ability to derive insights from complex datasets and contribute to impactful publications & managerial decision-making. Seeking leadership roles in public health/medical analytics.

PROJECT EXPERIENCE

Data Quality Overhaul for CDC-Sponsored Surveillance Study

July 2024 – December 2024

- Co-led initiative to revise data quality workflow in study with 3,000+ annual enrollees.
- Resulted in resolution of over 2,500 flags catching missing or mis-entered data.
- Overall reduction in CDC-prompted queries of local database, previously thousands of queries were triggered and now fewer than 200 monthly queries are anticipated.

Dashboard Design in Power BI

November 2024 – January 2025

- Created dashboards to monitor live enrollment and track virus circulation patterns in pediatric research study with 100+ laboratory results per week.

Electronic Consent Form Creation and Validation

August 2024 – Present

- Utilizing REDCap's e-consent framework improve compliance with institutional requirements.

WORK EXPERIENCE

Data Scientist

University of Pittsburgh, Department of Pediatrics – Pittsburgh, PA

May 2024 – Present

- Conduct statistical analyses with biomedical datasets to support pediatric research initiatives.
- Collaborate with cross-functional clinical teams to design studies and interpret results.
- Develop reproducible data pipelines and dashboards for ongoing studies.
- Present findings to stakeholders and contribute to publications.

Graduate Research Assistant

Colorado School of Public Health, Department of Biostatistics – Denver, CO

August 2021 – December 2023

- Led data analysis for public health research projects focused on pulmonology and health outcomes.
- Applied statistical methods and R/Python programming to clean, analyze, and visualize data.
- Co-authored peer-reviewed publications and presented at academic conferences.

Data Analyst

University of Pennsylvania, Department of Radiology – Philadelphia, PA

March 2018 – August 2020

- Analyzed imaging and clinical data to support research in radiology and neuroscience.
- Built Python software to streamline combination of diverse neuroimaging datasets.

- Collaborated with radiologists and researchers to identify lifespan trends and improve data quality.
- Contributed to the development of predictive models of age, dementia for clinical diagnostics.

EDUCATION

Master of Science in Biostatistics

University of Colorado Anschutz Medical Campus – Aurora, CO

Graduated: May 2023

Bachelor of Science in Economics

University of Pennsylvania, Wharton School – Philadelphia, PA

Graduated: May 2018

SKILLS

- Statistical Programming: R, SAS, Python
- Data Manipulation & Analysis: Regression, longitudinal modeling, survival analysis
- Scripting & Automation: Unix Bash, Microsoft Power Automate
- Data Visualization: ggplot2, shiny, matplotlib, Power BI
- Soft Skills: Collaborative research, scientific communication, problem-solving

SELECTED PUBLICATIONS

Pomponio, R., et al. (2020). Harmonization of large MRI datasets for the analysis of brain imaging patterns throughout the lifespan. *Neuroimage*, 208(1).

Habes, M., et al. (2020). The Brain Chart of Aging: Machine-learning analytics reveals links between brain aging, white matter disease, amyloid burden, and cognition in the iSTAGING consortium of 10,216 harmonized MR scans. *Alzheimer's & Dementia*, 17(1), 89-102.

Nasrallah, I., et al. (2021). Association of Intensive vs Standard Blood Pressure Control With Magnetic Resonance Imaging Biomarkers of Alzheimer Disease: Secondary Analysis of the SPRINT MIND Randomized Trial. *JAMA Neurology*, 78(5), 568-577.

Chen, A., et al. (2022). Harmonizing functional connectivity reduces scanner effects in community detection. *Neuroimage*, 256(1).

Dunbar, P., et al. (2024). Analgesia and Sedation Use During Noninvasive Ventilation for Acute Respiratory Failure. *Critical Care Medicine*, 52(7), 1043-1053.

Burnham, E., et al. (2024). Prevalence of Alcohol Use Characterized by Phosphatidylethanol in Patients With Respiratory Failure Before and During the COVID-19 Pandemic. *CHEST Critical Care*, 2(1).

Smith, J., et al. (2024). Donor derived cell free DNA in lung transplant recipients rises in setting of allograft instability. *Frontiers in Transplantation*, 3.

Held, N., et al. (2024). Extraneous load, patient census, and patient acuity correlate with cognitive load during ICU rounds. *Chest*, 165(6), 1448-1457.

Pomponio, R., et al. (2025). Phosphatidylethanol measures in patients with severe COVID-19-associated respiratory failure identify a subset with alcohol misuse. *Alcohol: Clinical & Experimental Research*, 49(1), 165-174.

OTHER RELEVANT EXPERIENCE

Peer reviewer for 5 journal articles (2020-present), Graduate teaching assistant (Fall 2023)

Maintainer of 'neuroHarmonize' (2021): software package available on Python Packaging Index (PyPI)